

Arithmetic & Number Sense

ANSWER KEY



Order of operations

- 1 Evaluate $12 + 6 \times (9 - 5) \div 3$.
Brackets first: $9 - 5 = 4$. Then $6 \times 4 = 24$, and $24 \div 3 = 8$. Finally $12 + 8 = 20$.
 - 2 Evaluate $8 + 2 \times 5^2 - 30 \div 6$.
Powers first: $5^2 = 25$. Then $2 \times 25 = 50$ and $30 \div 6 = 5$. So $8 + 50 - 5 = 53$.
 - 3 Evaluate $\frac{18 - 3 \times 4}{2} + 7$.
Numerator: $18 - 3 \times 4 = 18 - 12 = 6$. Then $6 \div 2 = 3$, and $3 + 7 = 10$.
 - 4 Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.
 $(7 - 2)^2 = 5^2 = 25$ and $3 \times (6 - 4) = 3 \times 2 = 6$. So $25 - 6 = 19$.
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Powers and roots

- 5 Evaluate 3^4 .
 $3^4 = 3 \times 3 \times 3 \times 3 = 81$.
 - 6 Evaluate $\sqrt{144}$.
 $12 \times 12 = 144$, so $\sqrt{144} = 12$.
 - 7 Evaluate $2^3 \times 2^2$ using the laws of exponents.
 $2^3 \times 2^2 = 2^{3+2} = 2^5 = 32$.
 - 8 Evaluate $\sqrt{81} + \sqrt{16}$.
 $\sqrt{81} = 9$ and $\sqrt{16} = 4$, so the sum is $9 + 4 = 13$.
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Working with fractions

- 9 Compute $\frac{3}{4} + \frac{5}{6}$.
LCD = 12: $\frac{3}{4} + \frac{5}{6} = \frac{10}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$.
- 10 Compute $\frac{7}{8} - \frac{1}{3}$.
LCD = 24: $\frac{7}{8} - \frac{1}{3} = \frac{21}{24} - \frac{8}{24} = \frac{13}{24}$.
- 11 Compute $\frac{2}{3} \times \frac{9}{10}$.
 $\frac{2 \times 9}{3 \times 10} = \frac{18}{30} = \frac{3}{5}$.

- 12 Compute $\frac{5}{6} \div \frac{5}{9}$.
Multiply by the reciprocal: $\frac{5}{6} \times \frac{9}{5} = \frac{45}{30} = \frac{3}{2}$.
- 13 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?
Cross-multiply: $5 \times 5 = 25$ and $4 \times 7 = 28$. Since $28 > 25$, $\frac{4}{5}$ is larger.
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Decimals

- 14 Compute $3.45 + 2.7$.
Align decimal points: $3.45 + 2.70 = 6.15$.
- 15 Compute $8.6 - 3.25$.
Align decimal points: $8.60 - 3.25 = 5.35$.
- 16 Compute 1.2×0.5 .
 $12 \times 5 = 60$, and there are 2 total decimal places, so $1.2 \times 0.5 = 0.60 = 0.6$.
- 17 Compute $7.5 \div 0.25$.
Multiply both by 100: $750 \div 25 = 30$.
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Place value and order of magnitude

- 18 Write 4,500,000 in the form $a \times 10^n$ with $1 \leq a < 10$.
 $4,500,000 = 4.5 \times 10^6$.
- 19 Which is greater, 6×10^4 or 58,000?
 $6 \times 10^4 = 60,000$, and $60,000 > 58,000$, so 6×10^4 is greater.
- 20 Order from least to greatest: 0.6 , $\frac{2}{3}$, 0.65 .
 $\frac{2}{3} \approx 0.667$. Ordered: $0.6 < 0.65 < \frac{2}{3}$.
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Mixed numbers and improper fractions

- 21 Convert $4\frac{2}{5}$ to an improper fraction.
 $4\frac{2}{5} = \frac{4 \times 5 + 2}{5} = \frac{22}{5}$.
- 22 Convert $\frac{47}{6}$ to a mixed number.
 $47 \div 6 = 7$ remainder 5, so $\frac{47}{6} = 7\frac{5}{6}$.
- 23 Compute $2\frac{1}{3} + 1\frac{3}{4}$.
As improper fractions: $\frac{7}{3} + \frac{7}{4}$. LCD = 12: $\frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}$.

- 24 Compute $5\frac{1}{2} - 2\frac{2}{3}$.
As improper fractions: $\frac{11}{2} - \frac{8}{3}$. LCD = 6: $\frac{33}{6} - \frac{16}{6} = \frac{17}{6} = 2\frac{5}{6}$.
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Absolute value and signed numbers

- 25 Evaluate $|-7| + |-3|$.
 $|-7| = 7$ and $|-3| = 3$, so the sum is $7 + 3 = 10$.
- 26 Evaluate $-8 + 5 \times (-2)$.
 $5 \times (-2) = -10$, so $-8 + (-10) = -18$.
- 27 Evaluate $|4 - 9| - |2 - 6|$.
 $|4 - 9| = |-5| = 5$ and $|2 - 6| = |-4| = 4$, so the result is $5 - 4 = 1$.
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Factors, multiples, GCD and LCM

- 28 Find the greatest common divisor (GCD) of 48 and 60.
 $48 = 2^4 \times 3$ and $60 = 2^2 \times 3 \times 5$. Common factors: $2^2 \times 3 = 12$.
- 29 Find the least common multiple (LCM) of 8 and 12.
 $8 = 2^3$, $12 = 2^2 \times 3$. LCM = $2^3 \times 3 = 24$.
- 30 List all the factors of 36, and state how many there are.
1, 2, 3, 4, 6, 9, 12, 18, 36 — that is 9 factors.
- 31 Is 91 a prime number? Justify your answer.
 $91 = 7 \times 13$, so it has factors other than 1 and itself. 91 is not prime.
- 32 Find the greatest common factor of 84 and 126.
 $84 = 2^2 \times 3 \times 7$, $126 = 2 \times 3^2 \times 7$. Common factors: $2 \times 3 \times 7 = 42$.
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Divisibility rules

- 33 Without dividing, determine whether 4,578 is divisible by 3.
Digit sum: $4 + 5 + 7 + 8 = 24$, which is divisible by 3 (since $24 = 3 \times 8$). So 4,578 is divisible by 3.
- 34 Is 2,340 divisible by 9?
Digit sum: $2 + 3 + 4 + 0 = 9$, which is divisible by 9. So 2,340 is divisible by 9.
- 35 Is 715 divisible by 5?
A number is divisible by 5 if it ends in 0 or 5. Since 715 ends in 5, it is divisible by 5.

- 36** Is 1,244 divisible by 4?

Check the last two digits: $44 \div 4 = 11$, an integer, so 1,244 is divisible by 4.

Estimation and rounding

- 37** Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.

Rounded: $200 \times 50 = 10,000$. Exact: $198 \times 51 = 10,098$ — close to the estimate.

- 38** Round 2.4863 to two decimal places.

The third decimal digit is 6, so round up: 2.49.

- 39** Estimate $\sqrt{50}$ to the nearest whole number.

$7^2 = 49$ and $8^2 = 64$; since 50 is closest to 49, the estimate is 7 (actual value ≈ 7.07).

- 40** A shipment weighs approximately 2,850 kg. Round this to the nearest hundred kilograms.

The tens digit is 5, so round up: 2,900 kg.

Number properties

- 41** If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?

Write $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, which is always even.

- 42** Find the remainder when 47 is divided by 6.

$6 \times 7 = 42$, and $47 - 42 = 5$. The remainder is 5.

- 43** Is 637 divisible by 7? Justify your answer.

$7 \times 91 = 637$, so yes, 637 is divisible by 7.

- 44** List all prime numbers between 30 and 50.

Checking each: 31, 37, 41, 43, 47 are prime (33, 35, 39, 45, 49 are not). That is 5 primes.

- 45** If the sum of two integers is odd, is their product always even, always odd, or can it be either?

A sum is odd only when one integer is even and the other is odd. Even $\times$ odd is always even, so the product is always even.

Word problems

- 46** A school orders 158 notebooks to be shared as equally as possible among 12 classrooms. How many notebooks does each classroom receive, and how many are left over?

$158 \div 12 = 13$ remainder 2 (since $12 \times 13 = 156$). Each classroom receives 13 notebooks, with 2 left over.

- 47** A caterer must seat 250 guests at tables that each hold 8 people. What is the minimum number of tables required?
 $250 \div 8 = 31.25$. Since a partial table still needs a full table, round up: 32 tables.
- 48** Three friends share a restaurant bill of 186 riyals equally, then each adds a 10-riyal tip. How much does each person pay in total?
Each share of the bill: $186 \div 3 = 62$ riyals. Adding the tip: $62 + 10 = 72$ riyals per person.
- 49** A red light flashes every 8 seconds and a green light flashes every 12 seconds. If both flash together right now, after how many seconds will they next flash together at the same time?
The lights next align after the LCM of 8 and 12, which is 24 seconds.
- 50** A water tank holds 480 liters and is filled by a pump running at 15 liters per minute. How long does it take to fill the tank from empty?
Time = $480 \div 15 = 32$ minutes.
- 51** A charity collects 2,340 riyals from 45 equal donations. How much was each donation?
Each donation = $2,340 \div 45 = 52$ riyals.